

Review:

Factor each expression completely.

Key

1.) $w^2 + 9w - 22$
 $(w+11)(w-2)$

2.) $4xy^2 + 24x^2y^6$
 $4xy^2(1+6xy^5)$

3.) $x^2 - 19x + 84$
 $(x-7)(x-12)$

4.) $-4x(x+3) + 3x^2(x+3)$
 $(-4x+3x^2)(x+3)$
 $x(3x-4)(x+3)$

5.) $6y^2 + 18$
 $6(y^2+3)$

6.) $n^2 - 19n + 90$
 $(n-9)(n-10)$

7.) $k^2 - 13k + 42$
 $(k-7)(k-6)$

18.) $144 - w^2$
 $(12-w)(12+w)$

9.) $m^2 - 6m + 5$
 $(m-5)(m-1)$

10.) $x^2 - x - 30$
 $(x+5)(x-6)$

11.) $14c^2d + 2cd^2$
 $2cd(7c+d)$

10.) $z^2 + 8z + 7$
 $(z+1)(z+7)$

13.) $24x + 48y$
 $24(x+2y)$

14.) $6f^4 + 18f^3 - 54f^2 - 162f$
 $6f(f^3 + 3f^2 - 9f - 27)$
 $6f(f^2(f+3) - 9(f+3))$
 $6f(f^2-9)(f+3) \rightarrow 6f(f-3)(f+3)^2$
 diff of squares

15.) $a^2 - b^2$
 $(a+b)(a-b)$

Factor each expression completely.

16.) $5q^3 + 7q^2 + 30q + 42$
 $q^2(5q+7) + 6(5q+7)$
 $(q^2+6)(5q+7)$

17.) $12d^4 + 12d^3 - 20d^2 - 20d$
 $4d(3d^3 + 3d^2 - 5d - 5)$
 $4d[3d^2(d+1) - 5(d+1)]$
 $4d(3d^2-5)(d+1)$

Factor each expression completely.

$$18.) 2x^3 + 8x^2 + 3x + 12$$

$$2x^2(x+4) + 3(x+4)$$

$$(2x^2+3)(x+4)$$

$$19.) 2x(x-5) + 3(x-5) + (x-5)$$

$$(2x+3+1)(x-5)$$

$$(2x+4)(x-5)$$

$$2(x+2)(x-5)$$

$$20.) (x-2)(6-4x) + (5x+4)(x-2)$$

$$(x-2)(6-4x+5x+4)$$

$$(x-2)(x+10)$$

$$21.) 81 - \sqrt{100c^2}$$

$$(9-10c)(9+10c)$$

$$22.) 2f^3 - 2f$$

$$2f(f^2-1)$$

$$2f(f+1)(f-1)$$

$$23.) 3k^2 + 14k - 80$$

$$\boxed{3} k^2 + 14k - 240$$

$$(k+24)(k-10)$$

$$(k+\frac{24}{3})(k-\frac{10}{3})$$

$$(k+8)(3k-10)$$

$$24.) w^2 + 5w - 6$$

$$(w+6)(w-1)$$

$$25.) 6p^2 - 29p + 28$$

$$\boxed{12} p^2 - 29p + 168$$

$$(p-8)(p-21)$$

$$(p-\frac{21}{6})(p-\frac{8}{6})$$

$$(p-\frac{7}{2})(p-\frac{4}{3})$$

$$(3p-4)(2p-7)$$

$$26.) d^8 - d^2$$

$$d^2(d^6-1)$$

$$27.) 2x^2 - 13x - 24$$

$$\boxed{2} x^2 - 13x - 48$$

$$(x-16)(x+3)$$

$$(x-\frac{16}{2})(x+\frac{3}{2})$$

$$(x-8)(x+\frac{3}{2})$$

$$(x-8)(2x+3)$$

$$28.) 3x^3 - 75x$$

$$3x(x^2-25)$$

$$3x(x-5)(x+5)$$

$$29.) x^2 + 17x + 70$$

$$(x+10)(x+7)$$